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26 November 1973
SUPERSEDING
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MILITARY SPECIFICATION

BOTTLE, SAFETY CAP, Plastic, Tablet and Capsule, Round, Light-Resistant

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers light-resistant, plastic bottles with child-resistant safety caps, for dispensing drugs in tablet or capsule form.

1.2 Classification.

* 1.2.1 Types. The safety caps and bottles shall be of the following types (see 6.1):

Type I	Press-lug lock cap
Type II	Child-resistant snap-on cap

1.2.2 Sizes. The bottles shall be of the following sizes:

Size	Vial capacity
a	4 dram (15 cc.)
b	7 dram (26 cc.)
c	12 dram (44 cc.)
d	16 dram (59 cc.)
e	20 dram (75 cc.)
f	30 dram (110 cc.)
g	40 dram (150 cc.)

2. APPLICABLE DOCUMENTS

* 2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

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Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

FOOD AND DRUG ADMINISTRATION

Federal Register Part 295.10	Testing Procedure for Special Packaging.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Child-resistance. Prior to delivery, the contractor shall submit proof that the safety closures under contract conform to the Poison Prevention Packaging Act of 1970 (Public Law 91-601 dated 30 December 1970) under FDA Regulation 295.10. Proof shall consist of a copy of data submitted to the Commissioner of Food and Drugs, on tests conducted in accordance with this FDAR.

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3.1.1 Compliance with above paragraph does not excuse compliance with all requirements of the contract and specification.

3.2 Material.

3.2.1 Bottle. The bottle shall be fabricated of rigid polystyrene plastic. The polystyrene plastic used in the fabrication of the bottles shall meet the requirements of the U.S.P. for Light Transmission for light-resistant containers.

3.2.2 Cap. The cap shall be fabricated of suitable polypropylene or polystyrene plastic. The plastic used shall be opaque to light.

3.3 Design and construction. The bottle shall be constructed so that, when the cap is affixed to the bottle, the removal of the cap resists the efforts of a child (see 3.1). The bottle shall have a one-piece closure with the child-resistant safety feature design incorporated in the closure cap and/or bottle as hereinafter indicated. The bottle shall be a vial-type straight-wall cylindrical container, except that the side walls may be slightly tapered. The side walls of the bottle shall be smooth. The bottle shall be free of rocking and wobbling when placed on a smooth, flat, and level surface. The closure design for cap and bottle shall conform to the requirements of either 3.3.1 or 3.3.2.

3.3.1 Press-lug lock cap design(Type I). The design of the bottle and closure cap shall be such that the cap snaps into a locked position by being rotated in a clockwise direction to stops or indents on the bottle to engage the locking safety feature. The locked cap shall be removable from the bottle without damage to either component. The release and removal of the closure cap shall be accomplished by two specific and coordinated actions: applying normal pressure to the top of the cap while simultaneously turning it in a counter-clockwise direction. The cylinder opening shall have six or more lugs on the exterior surface of the outer rim. The cap shall have vertical ridges on the outside rim to facilitate gripping for rotation, and shall have protrusions on the inside surface of the rim to mate with the lugs on the bottle. The interior of the cap shall have a flexible circular diaphragm or foamed liner for engagement with the open end of the bottle in closure. The diaphragm or liner shall be secured under the lugs to prevent it from being separated from the cap. The cap shall be capable of repeated operations of being secured to and removed from the bottle without damage to either component. When tested in accordance with 4.4.1, the safety closure shall function properly without evidence of the following:

a. Deterioration of the components on the bottle and/or cap, as a result of repeated use.

b. Removal of cap from safety closure position on bottle without applying normal (downward) pressure to top of cap.

c. Cap not removable from bottle.

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* 3.3.2 Child-resistant snap-on cap design (Type II). The safety closure design for the Type II bottle shall be as indicated in 3.3.2.1. The bottle shall be designed to accept a snap-on cap. The Type II bottle shall have a bead on the outer rim of the opening. The bead shall be of such dimension that it can accept and retain securely a snap-on cap. The supplied cap shall have a bead (or beads) around its inner rim so that when the cap engages the bottle, the bead of the cap falls below the bead of the bottle and hugs the surface of the bottle. The Type II bottle shall also have a flange positioned under the closure cap so that an obstruction is offered to attempts to pry the bottom edge of the cap so that it stretches in order to be lifted over the beaded rim of the bottle.

3.3.2.1 Index alignment lifting tab. The design of the bottle shall be such that the cap can be applied to the bottle in any position, but cannot be removed from the bottle unless a lifting tab on the rim of the cap is properly aligned with an index located on the bottle. The closure cap shall be capable of being rotated in any direction in order to have the lifting tab positioned with or away from the index on the bottle. The rim of the cap shall have a vertical ridge to facilitate gripping. The safety closure design shall be in accordance with one of the following:

a. A retention groove is formed on the exterior wall of the bottle between the beaded rim and flange (see 3.3.2). The retention groove shall trap a locking lug, which is part of the inner rim of the cap, when the lifting tab is not aligned with the index on the bottle. The lifting tab and locking lug shall be integrally molded to the rim of the cap. The alignment of the lifting tab with the index on the bottle shall position the locking lug below a notch located in the beaded rim of the bottle. The notch shall allow the locking lug (i.e. the cap) to be lifted off the bottle without obstruction from the rim. The index on the bottle shall be distinct and permanent and readily identifiable, and shall be located below the bottom of the closure cap.

b. The flange on the outside wall of the bottle (see 3.3.2) is of sufficient width to obstruct access to the lifting tab on the closure cap, so that the cap cannot be lifted off the beaded rim of the bottle. The lifting tab shall be integrally molded to the bottom rim of the cap. In closure the lifting tab shall project from the side of the cap, but shall not extend beyond the boundary of the flange in the safety closure position. The index on bottle shall be in the form of a notch or sector opening on the flange constructed so that, when the lifting tab is positioned over the notch, the lifting cap is accessible and can be used to lift the cap off the beaded rim of the bottle.

The cap shall be capable of repeated operations of being secured to and removed from the bottle without damage to either component. When tested in accordance with 4.4.2, the cap shall function properly without evidence of the following:

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a. Deterioration of the components on the bottle and/or cap as a result of repeated use.

b. Cap not adjustable to all positions on the bottle.

c. Cap able to be lifted off the bottle by gripping the rim of the cap.

d. Cap does not snap over or off the bead on the outer rim of the bottle.

e. Cap not removable from bottle when lifting tab is aligned with index on bottle.

f. Cap removable from bottle when lifting tab is not aligned with index on bottle.

3.3 Capacities and dimensions. The bottles shall be dimensioned so that the inside height is greater than the inside diameter. The bottles shall conform to Table I for capacity and tolerance:

Table I

Size	Capacity (drams)	Tolerance
a	4	+ 1.0/-0.5
b	7	+ 1.0/-0.5
c	12	+ 1.5/-0.5
d	16	+ 1.5/-1.0
e	20	+ 1.5/-0.5
f	30	+ 1.5/-0.7
g	40	+ 2.0/-0.5

3.4 Identification marking. Each bottle shall be permanently and legibly marked on the bottom with the name or registered trademark of the manufacturer and the capacity. Capacity shall be marked to the nearest dram. Instructions for removal of the cap from the bottle shall be permanently and legibly marked on each cap in contrasting color to the body of the cap.

3.5 Workmanship. Workmanship shall be first class throughout. The bottle and cap shall be supplied clean, dry and ready for use. Bottles and caps on all lots of material shall be interchangeable. Bottles and caps shall be free from defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, for the light-resistant characteristics of the bottle, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table II. The unit of product for inspection purposes shall be as indicated below.

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Table II Sampling for examination

	Unit of product	Inspection level	AQL (percent defective)
For visual examination			
Major A	Individual bottle & individual cap	II	2.5
Major B	Individual bottle	II	2.5
Minor A	Individual bottle & individual cap	II	4.0
Minor B	Individual bottle	II	4.0
For dimensional examination	Individual bottle	S-1	2.5

4.2.2 For testing. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table III. The unit of product for testing shall be the individual bottle and individual cap.

* Table III Sampling for tests

Product (see 1.2.1)	Characteristic	Requirement	Test procedure	Inspection level *	AQL
Type I	Safety closure durability	3.3.1	4.4.1	S-2	**
Type II	Safety closure durability	3.3.2.1	4.4.2	S-2	**

* Minimum sample size shall be 5.
** Acceptance number shall be 0.

4.3 Examination. The bottle and cap shall be examined in accordance with the defects listed in Table IV, but examination shall not be limited to defects listed.

* Table IV Classification of defects

Category	Defect
Major A	
101	Bottle or cap not free of cracks, chips and breaks.
102	Bottle or cap not free of deformation.
103	Bottle or cap not free of sharp edges. In addition to observation, cap shall be secured and opened in accordance with instructions (see Major 104 and 105). Cutting or scratching the hand or the feeling that the hand may be cut or scratched shall be considered evidence of a defect.
104	Cap cannot be secured to bottle.
105	Cap cannot be easily removed from bottle.
106	Bottle or cap not clean, dry or ready for use.
107	Cap cannot be adjusted to any desired position on bottle (Type II only)
108	Snap-on cap not secure and firm on bottle.
109	Bead of snap-on cap not below the bead of the bottle.
110	Bead of snap-on cap does not hug the surface of the bottle.

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Classification of defects (cont'd)

Category	Defect
Major B	
151	Bottle not free of crazing.
152	Bottle not stable on a smooth plane surface.
153	Marking not complete, correct or legible.
154	Bottle not free of colorless spots.
Minor A	
201	Bottle or cap not free of feathered edges.
202	Bottle or cap not free of burrs.
203	Bottle or cap not free of mars, marks or blemishes.
204	Bottle or cap not free of scratches which detract from its appearance.
Minor B	
251	Vertical walls of bottle not smooth.
252	Mold mark not free of sprues.

4.3.1 Dimensional examination. Bottles shall be examined for defects in dimensions. Capacity shall be determined by calculating from interior measurements. 3.7 cc/dram shall be used as the conversion factor from cubic centimeters to drams. As an alternate method, the capacity may be determined volumetrically.

4.4 Tests.

4.4.1 Safety closure durability (Type I only). The cap shall be locked in the safety closure position on the bottle (see 3.3.1). The locked cap shall then be released and removed from the bottle by using the method specified in 3.3.1. The procedure of locking the cap on the bottle and then removing the cap from the bottle shall be repeated twenty-five (25) times. Afterward, the cap shall again be locked in position on the bottle. With the cap in the closure position, it shall be determined whether the cap can be removed from the bottle by rotating the cap in the counter-clockwise direction without applying a downward (normal) pressure to the top of the cap.

★ 4.4.2 Safety closure durability (Type II only). The cap shall be applied to the bottle, and then released and removed from the bottle by using the method specified in 3.3.2.1. The procedure of applying the cap to the bottle and then removing the cap shall be repeated twenty-five (25) times. With the cap in the safety closure position on bottle, it shall be determined whether the cap can be removed from the bottle without aligning the lifting tab with the index on the bottle by using the following methods:

a. Applying upward pressure to lifting tab.
b. Gripping sides of cap and attempting to lift the cap off the bottle.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package containing 100 or 200 bottles and caps, of the type and size specified, constitutes one unit of issue (see Table V and 6.1).

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V Packaging quantities

Size	Capacity (drams)	Qty per package	Unit package	Intermediate package	Exterior container
a	4	200	1 unit	Not required	8 units
b	7	200	1 unit	Not required	6 units
c	12	200	1 unit	Not required	4 units
d	16	200	1 unit	Not required	4 units
e	20	200	1 unit	Not required	4 units
f	30	100	1 unit	Not required	4 units
g	40	100	1 unit	Not required	4 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

* 5.1.3.1 Unit package. 100 or 200 bottles and 100 or 200 caps of the same size shall be separately packaged in clean, transparent plastic bags of appropriate size having a wall thickness of not less than 2 mil. Closure shall be effected by heat sealing or by plastic-coated wire ties. Bags shall be further packaged in a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be secured to prevent opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirement of applicable table of PPP-B-636 shall apply for PPP-B-636 box.

* 5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table of PPP-B-636 shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

NOTE: The item identification shall be shown on the unit package and exterior container as follows: "BOTTLE, SAFETY CAP, Plastic, Tablet and Capsule, Round, Light-Resistant (insert applicable capacity and quantity)."

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6. NOTES

* 6.1 Ordering data. Procurement document should specify the following:

a. Title, number and date of this specification.
b. Type, size and capacity required (see 1.2).
c. National Stock Number (NSN).
d. Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades or sizes of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements. Suppliers of other types and/or classes of safety cap bottles conforming to the requirements of 3.1 may submit samples of their design and copies of their protocol to this procurement agency for consideration as to acceptability for current or future procurement. This procurement agency reserves the right to consult with other cognizant Military activities for a determination as to suitability of the design to meet Military requirements.

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
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DOCUMENT IDENTIFIER AND TITLE: MIL-B-36819B

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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